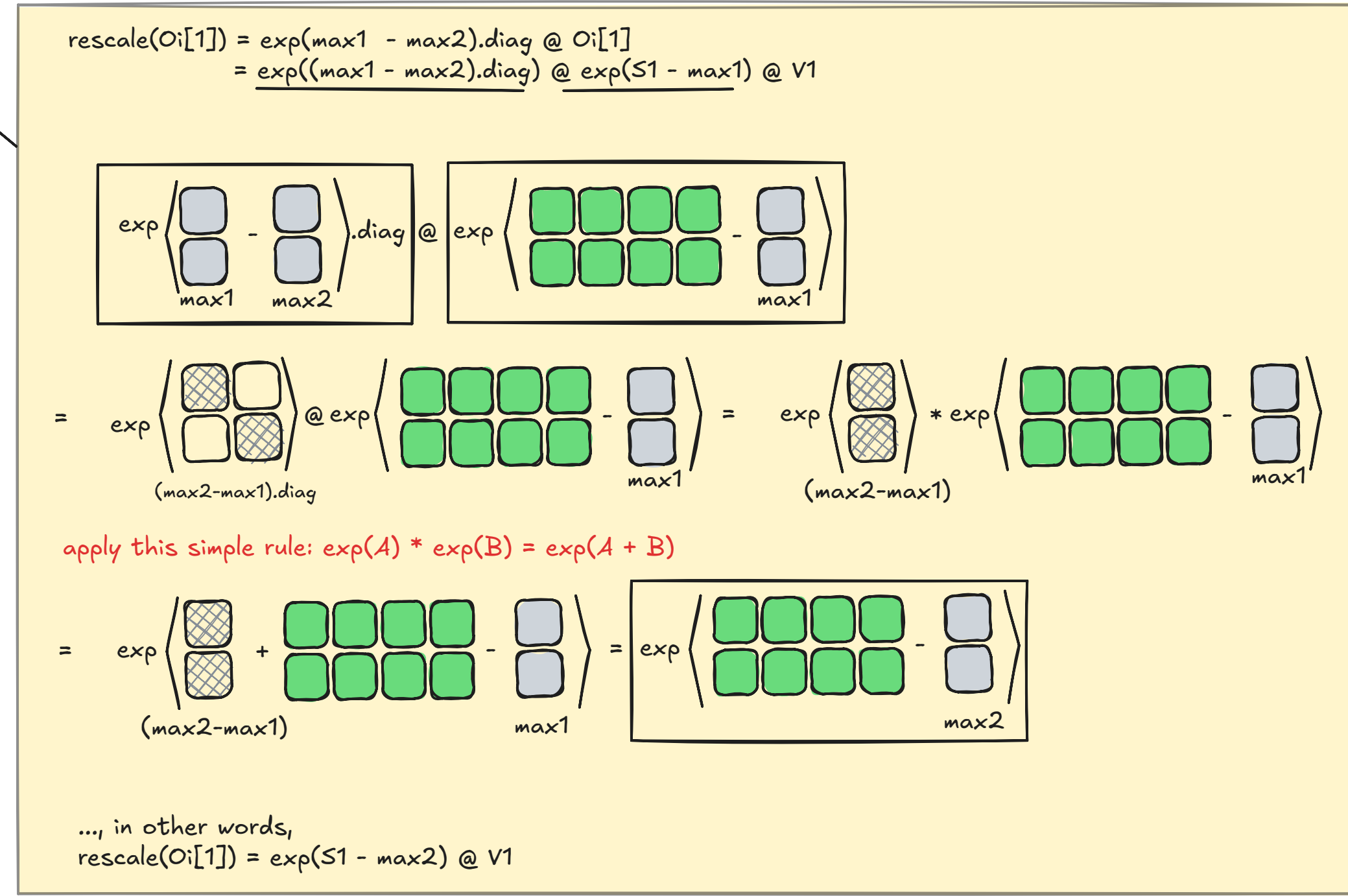
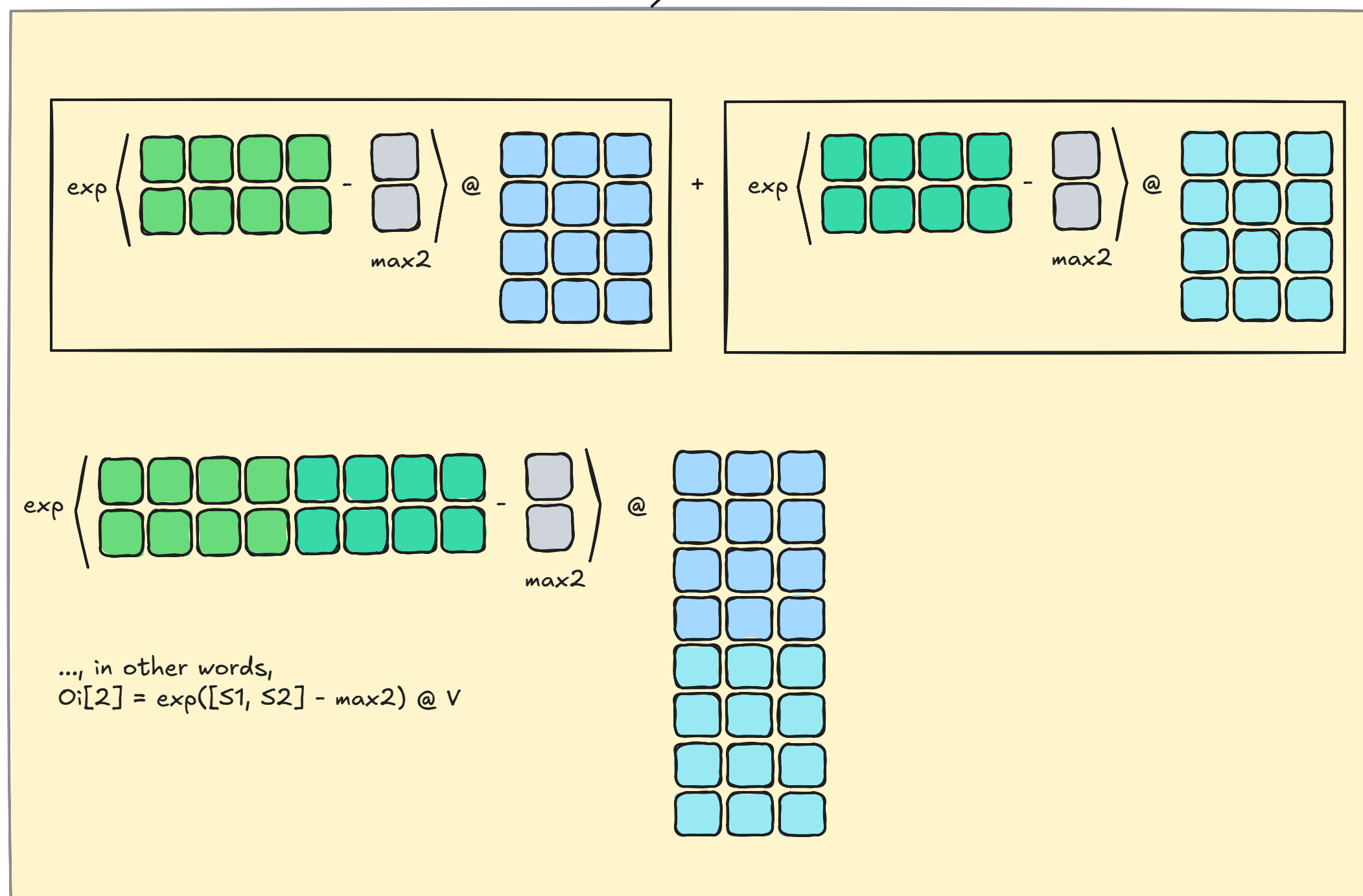
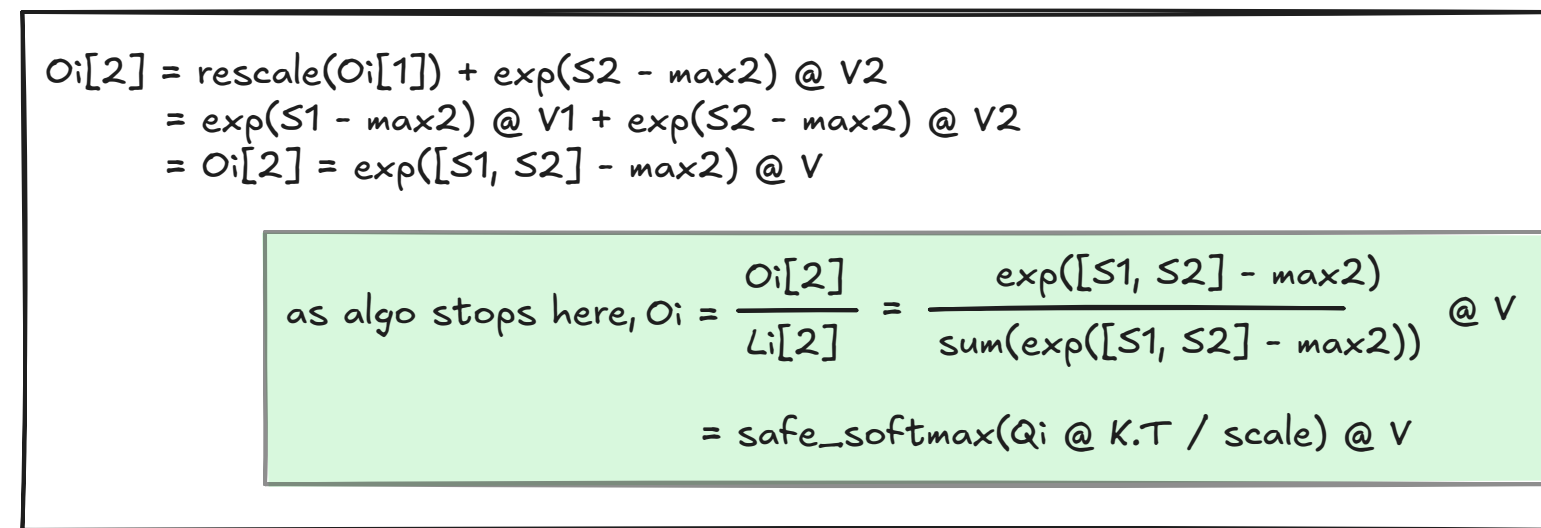
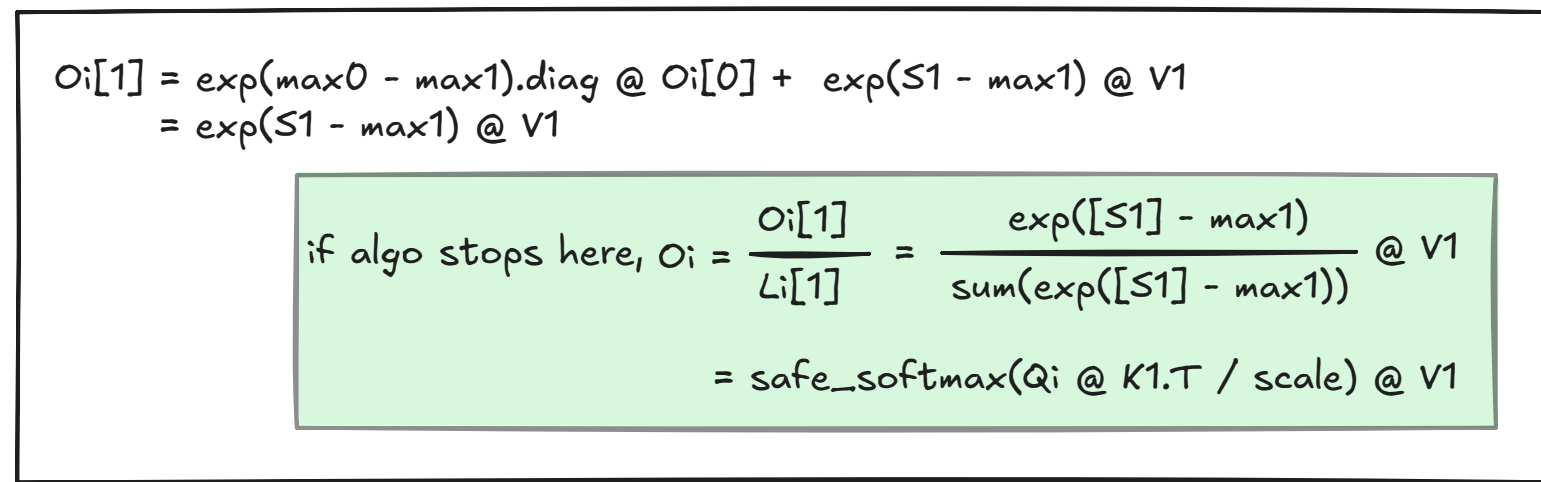
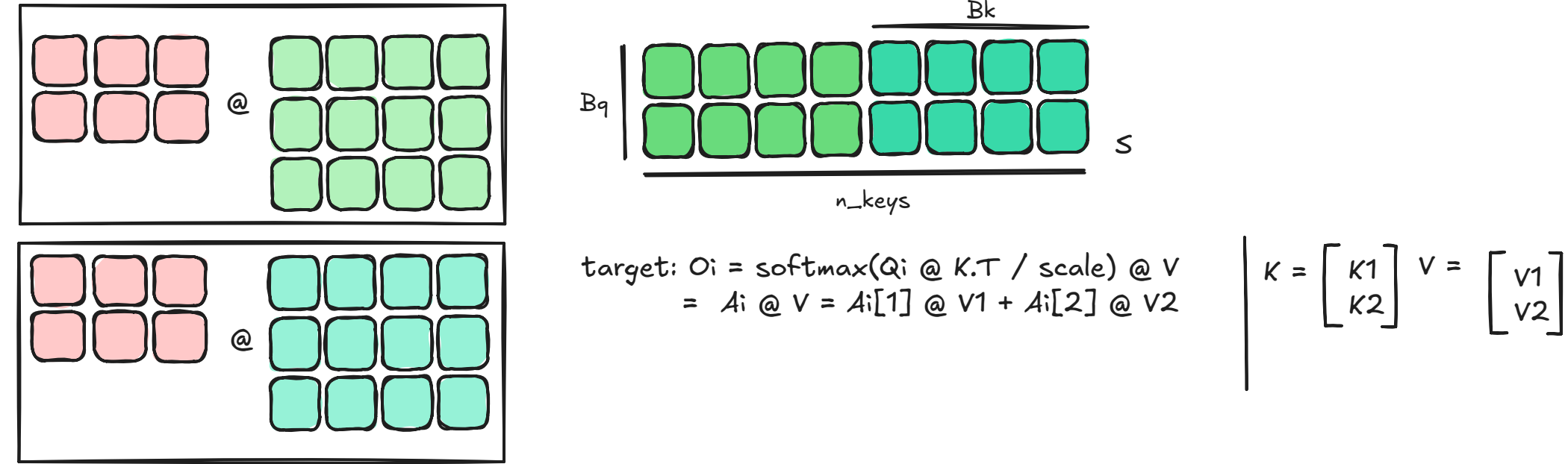
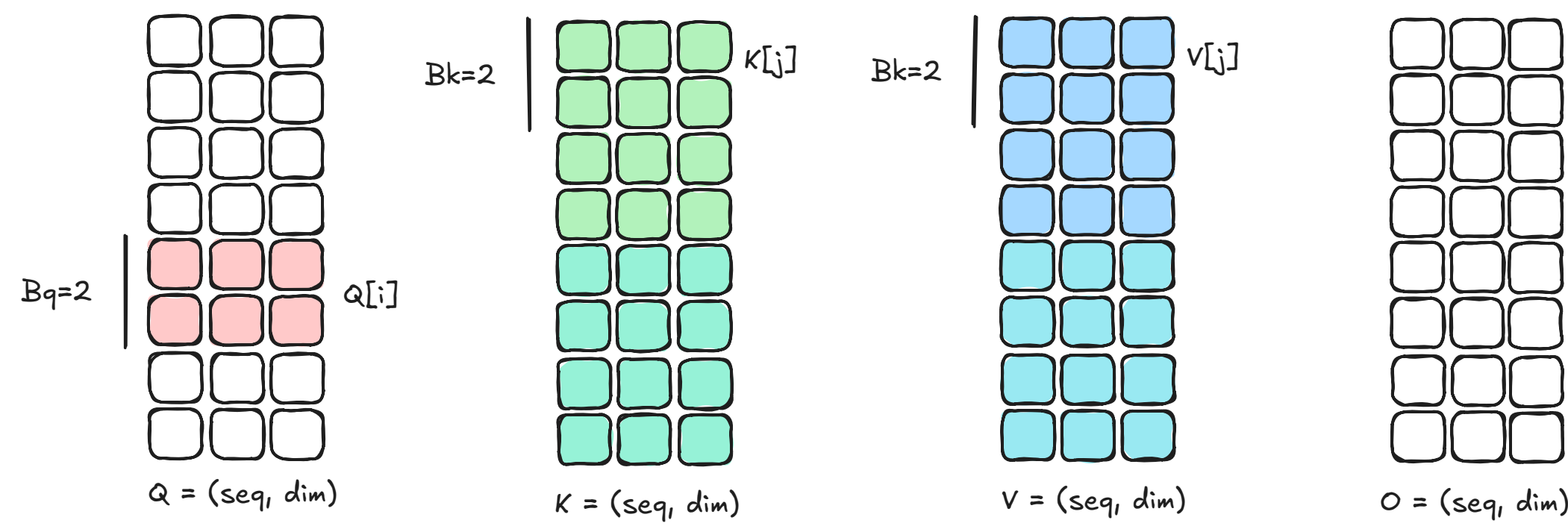
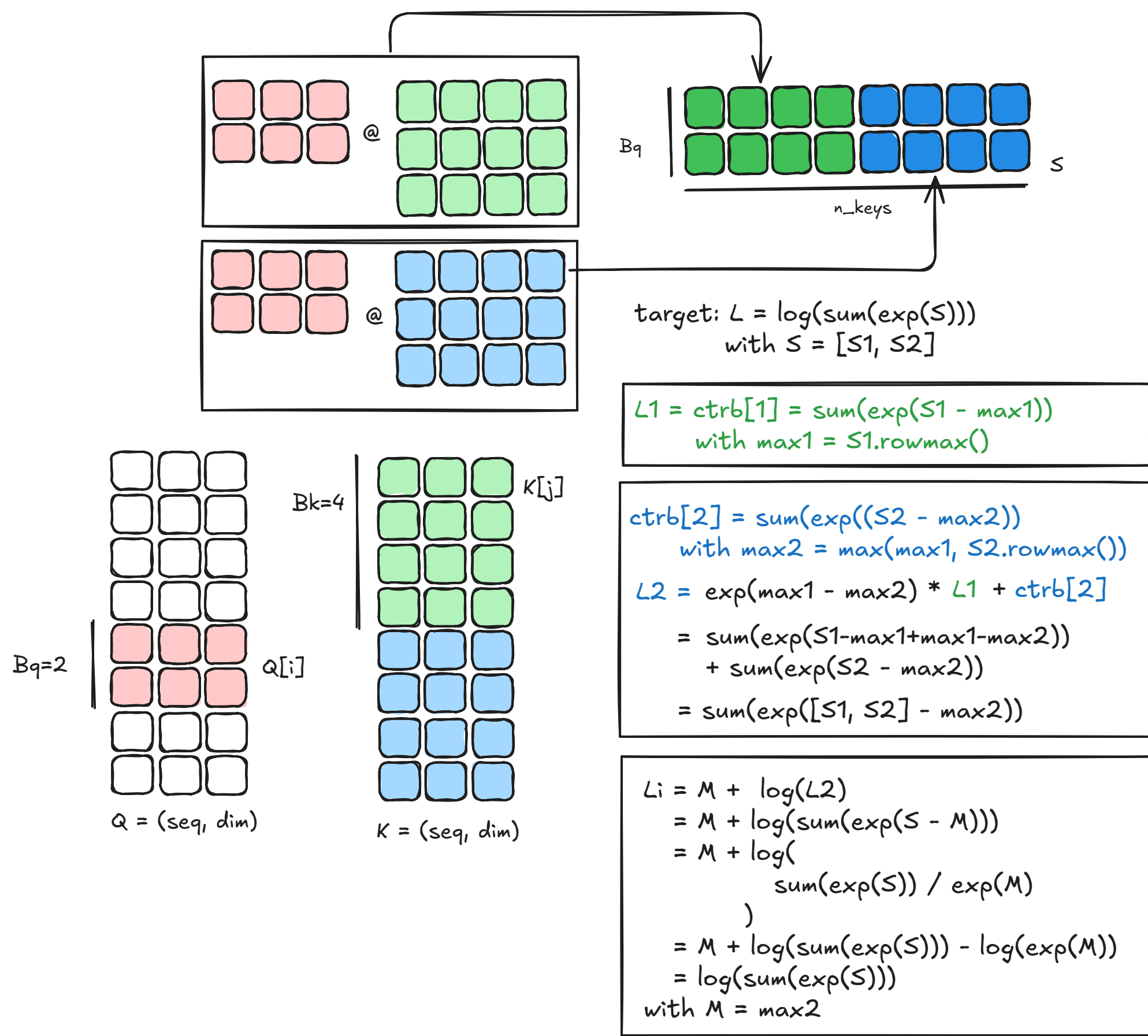




Algorithm 1 FlashAttention-2 forward pass

Require: $Q \in \mathbb{R}^{N_q \times d}$, $K, V \in \mathbb{R}^{N_k \times d}$, tile sizes B_q, B_k

Split Q into $T_q = \left\lceil \frac{N_q}{B_q} \right\rceil$ tiles $Q_1, \dots, Q_{T_q}$ of size $B_q \times d$

Split K, V into $T_k = \left\lceil \frac{N_k}{B_k} \right\rceil$ tiles $K^{(1)}, \dots, K^{(T_k)}$ and $V^{(1)}, \dots, V^{(T_k)}$ of size $B_k \times d$

for $i = 1, \dots, T_q$ **do**

Load Q_i from global memory

Initialize $O_i^{(0)} = \mathbf{0} \in \mathbb{R}^{B_q \times d}$, $l_i^{(0)} = 0 \in \mathbb{R}^{B_q}$, $m_i^{(0)} = -\infty \in \mathbb{R}^{B_q}$

for $j = 1, \dots, T_k$ **do**

Load $K^{(j)}, V^{(j)}$ from global memory

Compute tile of pre-softmax attention scores $S_i^{(j)} = \frac{Q_i(K^{(j)})^\top}{\sqrt{d}} \in \mathbb{R}^{B_q \times B_k}$

Compute $m_i^{(j)} = \max(m_i^{(j-1)}, \text{rowmax}(S_i^{(j)})) \in \mathbb{R}^{B_q}$

Compute $\tilde{P}_i^{(j)} = \exp(S_i^{(j)} - m_i^{(j)}) \in \mathbb{R}^{B_q \times B_k}$

Compute $l_i^{(j)} = \exp(m_i^{(j-1)} - m_i^{(j)}) l_i^{(j-1)} + \text{rowsum}(\tilde{P}_i^{(j)}) \in \mathbb{R}^{B_q}$

Compute $O_i^{(j)} = \text{diag}(\exp(m_i^{(j-1)} - m_i^{(j)})) O_i^{(j-1)} + \tilde{P}_i^{(j)} V^{(j)}$

end for

Compute $O_i = \text{diag}(l_i^{(T_k)})^{-1} O_i^{(T_k)}$

Compute $L_i = m_i^{(T_k)} + \log(l_i^{(T_k)})$

Write O_i to global memory as the i -th tile of O .

Write L_i to global memory as the i -th tile of L .

end for

Return the output O and the logsumexp L .

If I need to derive FlashAttention from scratch all over again,
I would start decompose the attention like above.